

SDTM DM DOMAIN PROGRAMMING USING SAS 9.4

Clinical SAS Portfolio Project

Prepared by

Riya Jagtap

B.Pharm | Clinical SAS / Clinical Data Management

Software: SAS 9.4 / SAS Studio

Standard: CDISC SDTM

Project Type: Learning / Portfolio Project

Note: This report uses sample clinical trial data for learning and portfolio purposes. No confidential patient or sponsor data is used.

1. Project Overview

This project was completed to practice the basic programming steps used while creating an SDTM Demographics (DM) domain. I used sample clinical trial datasets available in SAS Studio and worked mainly with Base SAS programming.

The work includes reviewing raw datasets, creating subject identifiers, deriving demographic and date variables, combining subject-level information, and checking the final DM dataset.

Main objective: Create a subject-level DM dataset in an SDTM-style structure from the available raw clinical trial data.

The project was done as a portfolio exercise to improve my understanding of Clinical SAS programming and SDTM data mapping.

2. Software and Standard Used

- SAS 9.4 / SAS Studio
- Base SAS programming
- CDISC SDTM DM-domain concepts
- Sample clinical trial datasets

3. Source Data Used

The programming used several raw datasets. The main source datasets reviewed during the project were:

Source dataset	Purpose in this project
RAW.DM	Subject demographic information such as age, sex, race, ethnicity and birth date.
RAW.IC	Informed consent date used for RFICDTC.
RAW.DS	Disposition information used for death and last-visit related variables.
RAW.EX	Exposure information used to derive reference exposure start/end dates.
RAW.DUMMY_RND	Treatment/randomization information used for ARM and ACTARM variables.
RAW.DAT_SUB	Subject status information used for screen-failure related variables.

4. DM Domain Structure

The DM domain is a subject-level domain. For this project, the final dataset was designed to contain one record for each subject and to bring together demographic, consent, disposition, exposure and treatment information.

The main identifiers used were STUDYID, DOMAIN, SITEID, SUBJID and USUBJID.

5. Creating Subject Identifiers

I created the study identifier and used the site and subject information from the raw data. SUBJID was derived from SUBNUM using SUBSTR. USUBJID was then created by joining STUDYID, SITEID and SUBJID with hyphens.

Example SAS logic:

```
STUDYID = "AAA-2022";  
DOMAIN = "DM";  
SITEID = SITENUM;  
SUBJID = SUBSTR(SUBNUM,4);  
USUBJID = CATX("-", STUDYID, SITEID, SUBJID);
```

Example result: AAA-2022-105-001.

6. Demographic Variable Derivation

The demographic variables were taken from RAW.DM. Temporary renamed variables were used for some source fields so that the SDTM variable names could be created clearly.

SDTM variable	Source / derivation
STUDYID	Assigned as AAA-2022 for this sample study.
DOMAIN	Assigned as DM.
SITEID	Derived from SITENUM.
SUBJID	Derived from SUBNUM using SUBSTR(SUBNUM,4).
USUBJID	CATX of STUDYID, SITEID and SUBJID.
BRTHDTC	BRTHDAT converted to ISO 8601 date using IS8601DA.
AGE	Mapped from source AGE.
AGEU	Assigned as YEARS.
SEX	Mapped from source SEX.
RACE	Mapped from source RACE.
ETHNIC	Mapped from source ETHNIC values.

7. Birth Date and ISO 8601 Conversion

BRTHDAT is stored as a SAS date in the raw DM dataset. I converted it to the character date format required for the SDTM-style output.

SAS logic:

```
BRTHDTC = PUT(BRTHDAT, IS8601DA.);
```

For example, a source date displayed as 22APR1975 is converted to 1975-04-22.

8. Ethnicity Mapping

The source ETHNIC variable contains coded values. I mapped the values used in the sample data to descriptive values in the final dataset.

```
if ETHNICX = "HISP" then ETHNIC = "HISPANIC OR LATINO";  
else if ETHNICX = "NHISP" then ETHNIC = "NOT HISPANIC OR LATINO";  
else if ETHNICX = "U" then ETHNIC = "UNKNOWN";  
else if ETHNICX = "DECLINED TO ANSWER" then ETHNIC = "NOT REPORTED";
```

A longer character length was assigned to ETHNIC so that the descriptive value would not be truncated.

9. Informed Consent Date (IC)

The RAW.IC dataset was used to derive RFICDTC. The source ICDAT value was converted to an ISO 8601 date.

```
RFICDTC = PUT(ICDAT, IS8601DA.);
```

This was then kept at the subject level using USUBJID.

10. Disposition Information

The RAW.DS dataset was reviewed for death date and last visit information. DTHDTC was derived from DSDTHDAT and RFPENDTC from DSLV DAT when the source dates were available.

```
if DSDTHDAT ne . then DTHDTC = PUT(DSDTHDAT, IS8601DA.);  
if DSDTHDAT ne . then DTHFL = "Y";  
if DSLV DAT ne . then RFPENDTC = PUT(DSLV DAT, IS8601DA.);
```

During the sample-data check, no non-missing death dates were present in the reviewed DS data. Therefore, a death date was not created artificially.

11. Exposure Data and Reference Dates

RAW.EX contains multiple exposure records for some subjects. For this project, only records where the study drug was taken (EXYN='Y') and the exposure date was available were used for reference exposure dates.

The exposure date and time were combined into an ISO-style date-time value:

```
RFXSTDTC = CATS(PUT(EXSTDAT,YYMMDD10.),'T',PUT(EXSTTIM,TOD8.));
```

The records were sorted by USUBJID and exposure date-time. The first exposure was used for the reference start variables and the last exposure was used for the reference end variables in this portfolio derivation.

This step is important because simply using NODUPKEY by USUBJID would remove additional exposure records instead of determining the first and last exposure.

12. Treatment Information

Treatment/randomization information was taken from RAW.DUMMY_RND. ARMCD and ACTARMCD were mapped from the treatment code, with ARM and ACTARM descriptions derived for the sample treatment values.

The treatment variables were then kept by USUBJID so that they could be merged into the subject-level DM dataset.

13. Screen Failure Information

RAW.DAT_SUB was reviewed for subject status. For the sample data, status information was used to identify subjects with a screen-failure status and derive the related subject-level variable.

14. Combining the Data

After the individual source datasets were processed, the subject-level datasets were combined using USUBJID. The demographic dataset was used as the base because the DM domain requires subject-level demographic information.

Basic merge approach:

```
DM1 + IC + DS + EX + TRT + SCF → FINAL
```

The final variable order was then controlled using RETAIN and the required output variables were kept.

15. Final DM Variable List

Variable	Role in final DM
STUDYID	Study identifier
DOMAIN	Domain identifier (DM)
USUBJID	Unique subject identifier
SUBJID	Subject identifier within site
RFSTDTC / RFENDTC	Reference start/end date-time
RFXSTDTC / RFXENDTC	Reference exposure start/end date-time
RFICDTC	Informed consent date-time
RFPENDTC	Reference period end date-time
DTHDTC / DTHFL	Death date and death flag when applicable
SITEID	Site identifier
BRTHDTC	Date of birth
AGE / AGEU	Age and age unit
SEX / RACE / ETHNIC	Demographic characteristics
ARMCD / ACTARMCD	Planned and actual treatment codes
ARM / ACTARM	Planned and actual treatment descriptions
ARMNRS / ACTARMUD	Reason/description fields used in the sample derivation
COUNTRY	Country value

16. Validation Checks

I used basic SAS checks to review the final dataset before considering it complete. The purpose was to identify duplicate subjects, missing key identifiers and unexpected values.

Examples of checks:

```
proc contents data=SDTM.DM varnum;  
run;
```

```
proc sort data=SDTM.DM out=DM_CHECK dupout=DM_DUP nodupkey;  
by USUBJID;  
run;
```

The duplicate check was treated as a validation step. NODUPKEY should not be used to silently remove records from the final DM dataset.

Other checks included reviewing the final dataset with PROC PRINT and checking the source records for missing exposure date/time values.

17. Example Final Output

The final DM output contains subject-level records with the derived identifiers, demographic variables, reference dates, treatment information and other applicable variables.

The actual SAS-generated output is stored separately in the GitHub repository under the Output folder. This report is the documentation for the programming work.

18. Problems Faced During Programming

While working on the project, I faced a few practical SAS programming issues. These were useful for understanding how source data can affect SDTM derivations.

- Character values can be truncated if the target variable length is too short. This occurred during the ETHNIC derivation and was handled by assigning a larger length.
- RAW.EX contained more than one record for some subjects. A simple NODUPKEY step would not be appropriate for deriving first and last exposure dates.
- Missing exposure dates had to be excluded from the first/last exposure derivation so that a missing record was not selected as the last record after sorting.
- The source datasets did not all have the same number of records per subject, so they needed to be reduced to subject-level information before the final merge.

19. Learning Outcomes

From this project I practiced:

- Reading and reviewing raw clinical datasets
- Using DATA steps and SET statements
- Character and date derivations
- SUBSTR and CATX/CATS functions
- BY-group processing and sorting
- Merging subject-level datasets
- Creating ISO 8601 date and date-time values
- Basic SDTM DM mapping
- Performing simple validation checks

20. Conclusion

This project helped me understand the basic flow from raw clinical trial data to an SDTM-style DM domain. I was able to practice subject identifier creation, demographic mapping, date derivations, exposure-based reference dates, treatment mapping and basic validation.

The project is intended as a beginner-level Clinical SAS portfolio project. The datasets are sample data and the derivations should not be treated as sponsor-specific production standards.